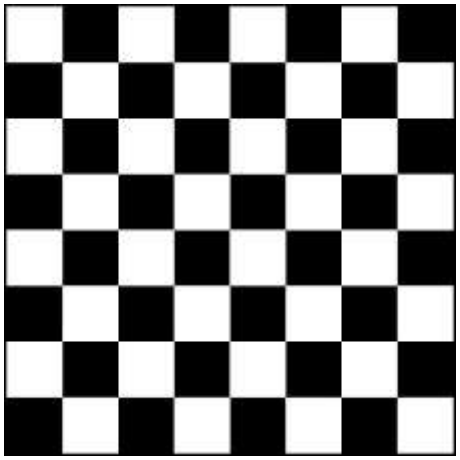
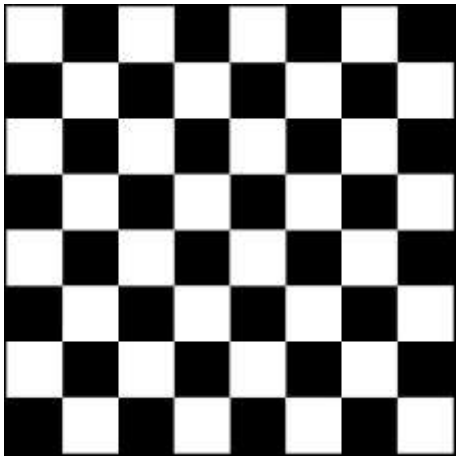
```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import Image
%matplotlib inline
```

```
Image(filename="/content/imgv.png")
```



```
Image(filename="/content/imgv.png")
```



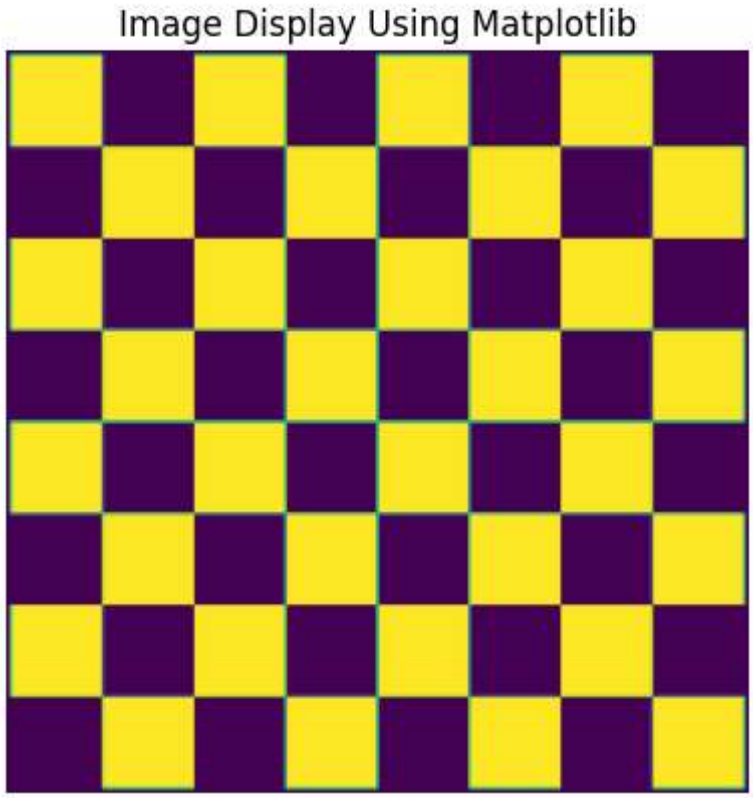
```
cb_img = cv2.imread("/content/imgv.png", cv2.IMREAD_GRAYSCALE)
print(cb_img)
```

```
[[  0   0   0 ...   0   0   0]
 [  0 138 191 ...   0   0   0]
 [  0 191 255 ...   0   0   0]
 ...
 [  0   0   0 ... 255 191   0]
 [  0   0   0 ... 191 138   0]
 [  0   0   0 ...   0   0   0]]
```

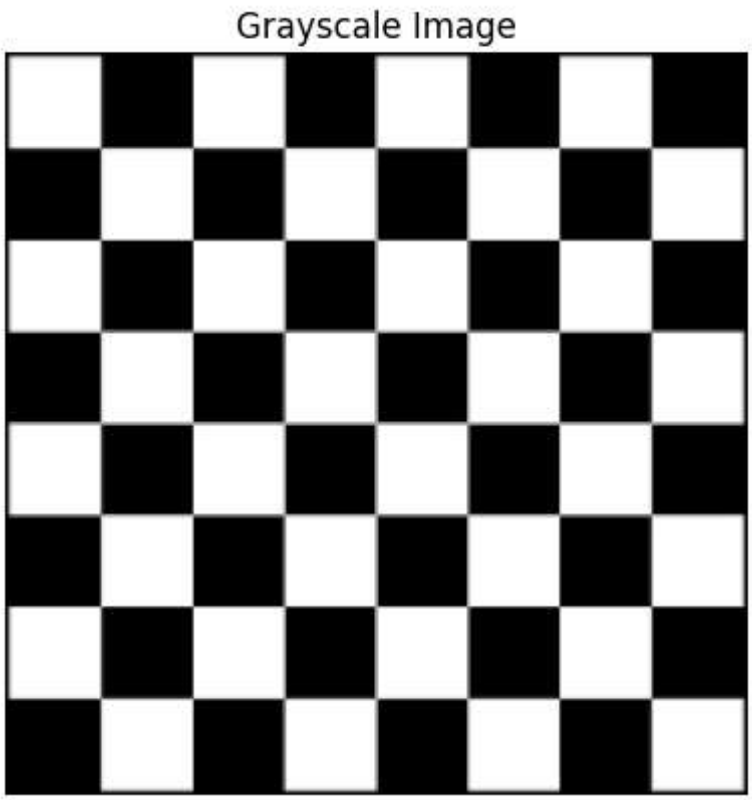
```
print("Shape of the image:", cb_img.shape)
print("Data type of the image:", cb_img.dtype)
print("Size of the image:", cb_img.size)
```

```
Shape of the image: (225, 225)
Data type of the image: uint8
Size of the image: 50625
```

```
plt.imshow(cb_img)
plt.title("Image Display Using Matplotlib")
plt.axis("off")
plt.show()
```




```
plt.imshow(cb_img, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
```




```
Image(filename="/content/avacado.jpg")
```

```
avacado_img = cv2.imread("/content/avacado.jpg", cv2.IMREAD_COLOR)

print("Image shape:", avacado_img.shape)
print("Image size:", avacado_img.size)
print("Image data type:", avacado_img.dtype)
```

```
Image shape: (180, 280, 3)
Image size: 151200
Image data type: uint8
```

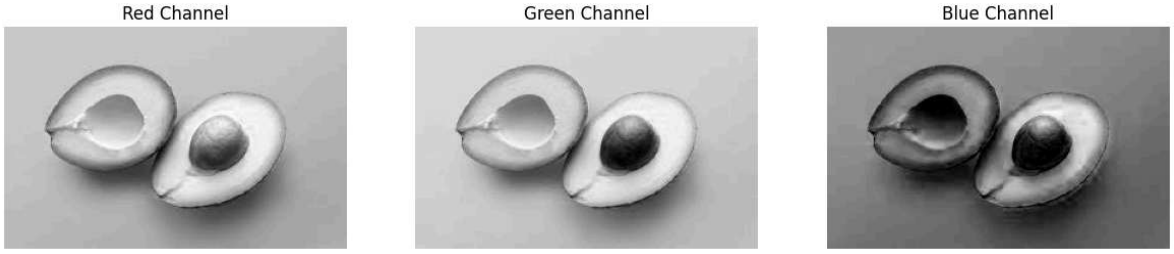
```
plt.imshow(avacado_img)
plt.axis("off")
plt.show()
```

```
avacado_img_rgb = avacado_img[:, :, ::-1]

plt.imshow(avacado_img_rgb)
plt.axis("off")
plt.show()
```

```
b, g, r = cv2.split(avacado_img)
```

```
plt.figure(figsize=(20,5))

plt.subplot(1,4,1)
plt.imshow(r, cmap="gray")
plt.title("Red Channel")
```

```
plt.axis("off")

plt.subplot(1,4,2)
plt.imshow(g, cmap="gray")
plt.title("Green Channel")
plt.axis("off")

plt.subplot(1,4,3)
plt.imshow(b, cmap="gray")
plt.title("Blue Channel")
plt.axis("off")

plt.show()
```




```
merged_img = cv2.merge((b, g, r))

plt.imshow(merged_img[:, :, ::-1])
plt.title("Merged Image")
plt.axis("off")
plt.show()
```

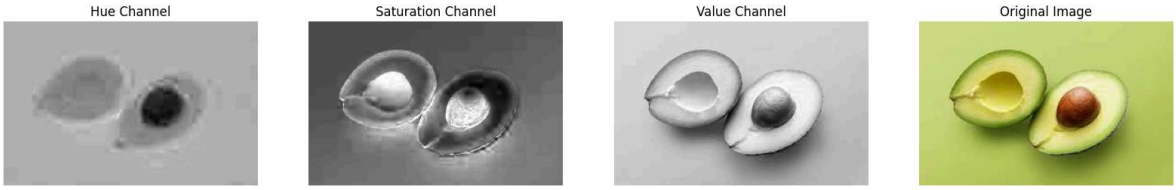
Merged Image

```
avacado_img_bgr = cv2.imread("/content/avacado.jpg")
avacado_img_rgb = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2RGB)

plt.imshow(avacado_img_rgb)
plt.title("BGR to RGB")
plt.axis("off")
plt.show()
```

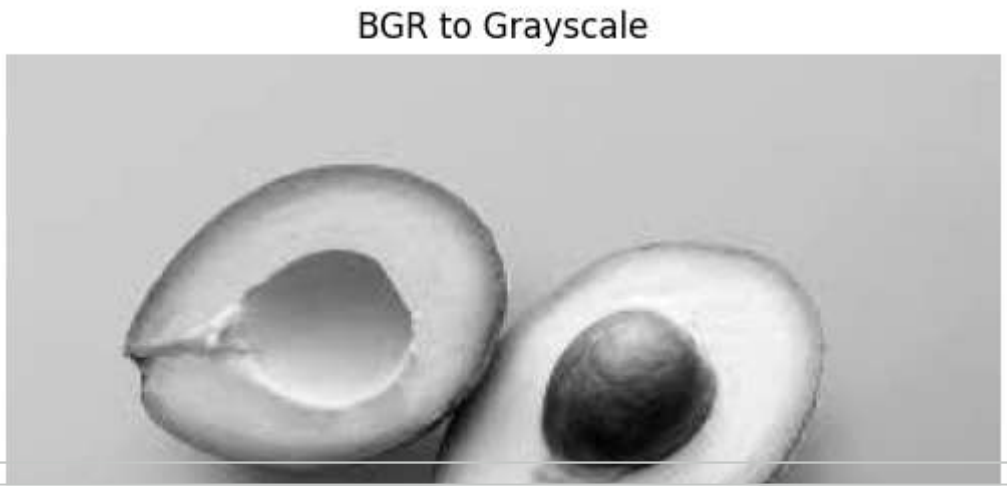
BGR to RGB

```
avacado_img_hsv = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2HSV)
h, s, v = cv2.split(avacado_img_hsv)
```

```
plt.figure(figsize=(20,5))

plt.subplot(1,4,1)
plt.imshow(h, cmap="gray")
plt.title("Hue Channel")
plt.axis("off")

plt.subplot(1,4,2)
plt.imshow(s, cmap="gray")
plt.title("Saturation Channel")
plt.axis("off")

plt.subplot(1,4,3)
plt.imshow(v, cmap="gray")
plt.title("Value Channel")
plt.axis("off")

plt.subplot(1,4,4)
plt.imshow(avacado_img_rgb)
plt.title("Original Image")
plt.axis("off")

plt.show()
```

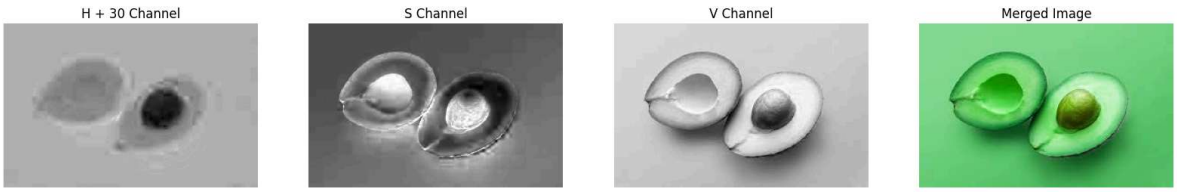



```
avacado_img_gray = cv2.cvtColor(avacado_img_bgr, cv2.COLOR_BGR2GRAY)

plt.imshow(avacado_img_gray, cmap="gray")
plt.title("BGR to Grayscale")
plt.axis("off")
plt.show()
```




```
h_new = h + 30
avacado_img_modified = cv2.merge((h_new, s, v))
avacado_img_rgb_modified = cv2.cvtColor(avacado_img_modified, cv2.COLOR_HSV2
```

```
h, s, v = cv2.split(avacado_img_modified)

plt.figure(figsize=(20,5))

plt.subplot(1,4,1)
plt.imshow(h, cmap="gray")
plt.title("H + 30 Channel")
plt.axis("off")

plt.subplot(1,4,2)
plt.imshow(s, cmap="gray")
plt.title("S Channel")
plt.axis("off")

plt.subplot(1,4,3)
plt.imshow(v, cmap="gray")
plt.title("V Channel")
plt.axis("off")

plt.subplot(1,4,4)
plt.imshow(avacado_img_rgb_modified)
plt.title("Merged Image")
plt.axis("off")

plt.show()
cv2.imwrite("avacado_saved.jpg", avacado_img_rgb_modified)
```




```
True
```